Type MS · Describe genetic modification and gene therapy for human health

Answer 1 of 4 · Question 6 part whole · 6 marks

2022 JUN · Paper 01 · find part whole in this mark scheme extract

6	Any six from the following: • remove plasmid from bacteria; • locate the human insulin gene • cut/remove insulin gene (from human); • cut open the plasmid; • using restriction enzymes; • add the human insulin gene to the plasmid; • use of ligase; • ref recombinant DNA; • return the plasmid to a bacterial cell;	Suitably drawn **and** annotated diagrams can score full marks.	6
			Total 6

Question number	Answer	Notes	Marks

Type MS · Describe genetic modification and gene therapy for human health

Answer 2 of 4 · Question 6 part c · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

Number					
6	(a)	(i)	Three from: • the use of embryonic stem cells is not needed/embryos not destroyed; • easily available source of cells/easy to obtain/no need to find a donor; • same cell/tissue/blood type of the patient/same antigens; • reduced chance of rejection/not recognised as foreign;		3
		(ii)	Two from: • unlicensed clinics could be used; • stem cells could develop into tumours/cancer; • risk of rejection (if the cells are not the recipients); • increased risk of infection;		2
	(b)	(i)	860 000 and 480 000; 380 000;	Ecf allow one mark for incorrectly derived figures from graph but correct answer from these figures	2
		(ii)	**D** structure Z;		1
	(c)		Three from: • cut out desired/beta-carotene/gene/gene for vitamin A; • using/cut with restriction enzymes; • insert into plasmid; • insert into/infect rice genome; • to produce golden rice;		3

Total 11 marks

Type MS · Describe genetic modification and gene therapy for human health

Answer 2 of 4 · Question 6 part c · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe genetic modification and gene therapy for human health

Answer 3 of 4 · Question 7 part e · 2 marks

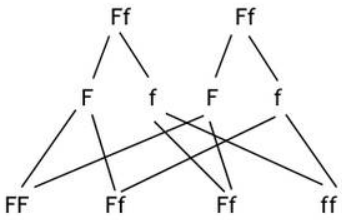
2025 JUN · Paper 02 (regional R) · find part e in this mark scheme extract

7	(a)	amylase;		1
	(b)	Any two from • enzymes are immobilised on testing strip; • react with/breaks down glucose (in urine); • (catalyse a chemical reaction that) causes a colour change (in strip);		2
	(c)	pancreas;		1
	(d)	• rejection/immune response/allergies; • (immune system) recognised it as foreign/antigen;		2
	(e)	any two from • use restriction enzymes; • cut out insulin gene from human DNA; • insert human insulin gene into yeast genome/plasmid/genetic material;		2
			Total for Question 7 = 8 marks	

Type MS · Describe genetic modification and gene therapy for human health

Answer 4 of 4 · Question 8 part b · 3 marks

2024 JUN · Paper 02 · find part b in this mark scheme extract

Number				
8	(a) (i)	**B** individual 3 The following are incorrect because: **A** individual 1 is heterozygous **C** Individual 5 is heterozygous **D** individual 3 is homozygous dominant		(1)
	(ii)	**D** 4 The following are incorrect because: **A** there are more than 1 heterozygous genotypes **B** there are more than 2 heterozygous genotypes **C** there are more than 3 heterozygous genotypes		(1)
	(iii)	• they have a dominant allele; • over-rides/ masks the effect of the recessive allele/ must have two recessive alleles for the condition to show/ must have homozygous recessive alleles ;	do not allow dominant allele overpowering the recessive allele	(2)
	(iv)	• heterozygous parent genotypes; • gametes; • genotypes of offspring; • offspring genotypes linked to the phenotype for the disease and showing that showing that 1 of the 4 genotype has the disease;  	mps 2,3 and 4 marks for correct Punnett square showing all details	(4)
	(b)	• reference to use of restriction enzyme • (to obtain) the normal/ non faulty/ dominant allele); • use of virus/ correctly named virus; • inserts allele into person **cells**; • functional/normal protein produced;	Accept liposomes ignore plasmids Penalise one for using genes instead of alleles.	(3)

Total for Question 8 = 11 marks